# Anatomy of an Interactive Lesson

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1 Bond | 2 Spring | 3 Real spring | 4 Energy well

**Concept scaffold** — each tab adds exactly one idea.

## 1 Learn the mathematics

build the model

**Background: two atoms, one invisible tug-of-war**

$F(r)$ = push apart if $r < r_0$ · zero at $r_0$
pull back — weakening to fracture — if $r > r_0$

WORKED EXAMPLE | KEY TAKEAWAYS

*Plain-language reasoning builds into the governing equation.*

**same idea, made tunable ↓**

## 2 Drive the simulation

test the model

CONTROLS

Separation 1.07 × $r_0$

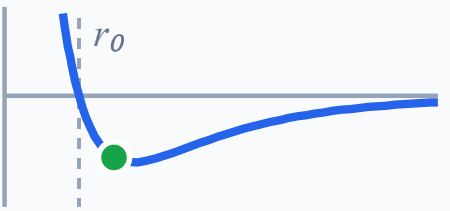
Bond strength 1.0×

Squeezed | Natural | Stretched | Break

THE TWO ATOMS



FORCE–SEPARATION




✓ **Correct** — the bond now pulls back past $r_0$.

*Move a slider — the picture and the self-check update live.*

**Verify-then-trust** — predict, then test by dragging.